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The Optimal Recovery of Smooth Functions

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Summary. It is shown that there is a positive lower bound, c, to the uniform error in any scheme designed to recover all functions of a certain smoothness from their values at a fixed finite set of points. This lower bound is essentially attained by interpolation at the points by splines with canonical knots. Estimates of c are also given.

Suppose $0 \leqq x_1 \leqq x_2 \leqq \cdots \leqq x_{n+r} \leqq 1$ with $x_i < x_{i+n}$, $n \geqq 1$ and $i = 1, \ldots, r$. Let

$$W_\infty^{(n)}[0,1] = \{f \in C^{n-1}[0,1]: \quad f^{(n-1)} \text{ abs. cont.}; \quad f^{(n)} \in L_\infty[0,1]\}.$$

Suppose $f \in W_\infty^{(n)}[0,1]$ and the values $f(x_1), \ldots, f(x_{n+r})$ are known to us, with the convention that if $x_{i-1} < x_i = x_{i+1} = \cdots = x_{i+s} < x_{i+s+1}$ then $f(x_{i+j}) = f^{(j)}(x_i)$, $j = 1, \ldots, s$. Let F denote the vector $(f(x_1), \ldots, f(x_{n+r}))$, then our aim is to determine how well it is possible to recover f, assumed to satisfy $\|f^{(n)}\| \leqq 1$, (where $\|\cdot\|$ denotes the (essential) sup norm) from F; where the measure of goodness of approximation to f is taken to be the sup norm in $C[0,1]$. In the first section we show that there is a positive lower bound to the error of any recovery scheme, and that perfect spline interpolation is a good recovery scheme. In the second section we present linear recovery schemes whose errors come arbitrarily close to the lower bound. The third section is devoted to various estimates of the intrinsic error.

Section 1

By a recovery scheme, S, we mean *any* transformation

$$S: \mathscr{F} \to W_\infty^{(n)}[0,1],$$

where $\mathscr{F}$ is the set of all vectors F: $(f(x_1), \ldots, f(x_{n+r}))$ and $f \in W_\infty^{(n)}[0,1]$ satisfies $\|f^{(n)}\| \leqq 1$.

Given S we consider the error

$$E_S(x) = E_S(x_1, \ldots, x_{n+r}) = E_S = \sup_{\substack{f \in W_\infty^{(n)}[0,1] \\ \|f^{(n)}\| \leqq 1}} \|f - SF\|$$

and

$$E(x) = E(x_1, \ldots, x_{n+r}) = E = \inf_S E_S.$$

An important technique for us is *perfect spline interpolation* as described by Karlin [6]. (See also de Boor [4].) Given F, according to Karlin [6] there exists a perfect spline of degree n

$$p(t) = \sum_{i=0}^{n-1} a_i t^i + d\left[t^n + 2\sum_{i=1}^{k-1} (-1)^i (t - \xi_i)_+^n\right] \tag{1}$$

with knots satisfying $0<\xi_1<\cdots<\xi_{k-1}<1$ and $k\leqq r$ such that, in our notation, $P=F$. Moreover if $g\in W_\infty^{(n)}[0,1]$ satisfies $G=P$ then

$$\|g^{(n)}\|\geqq\|p^{(n)}\|.$$

Choose $\xi\neq x_i$, $i=1,\ldots,n+r$ ($\xi\in[0,1]$) and consider the set of nodes obtained by adjoining ξ to x. Suppose $f_\xi(x_i)=0$, $i=1,\ldots,n+r$, $f_\xi(\xi)=1$, and $f_\xi\in W_\infty^{(n)}[0,1]$. Then Karlin's result implies that there exists a perfect spline of degree n ($\not\equiv 0$) having at most r knots and taking the value zero at $x_1,\ldots,x_{n+r}$. We need the following uniqueness result. (Cf. Cavaretta [2], Karlin [6]).

Lemma 1. If q is a perfect spline of degree n having at most r knots and satisfying

$$q(x_i)=0,\quad i=1,\ldots,n+r \tag{2}$$

and

$$\|q^{(n)}\|=1 \tag{3}$$

then it is unique (up to multiplication by -1).

Proof. It suffices to show that if $f\in W_\infty^{(n)}[0,1]$ satisfies $\|f^{(n)}\|\leqq 1$ and $f(x_i)=0$, $i=1,\ldots,n+r$ then $|f(t)|\leqq|q(t)|$, $0\leqq t\leqq 1$. To establish this inequality suppose that $|f(t_0)|>|q(t_0)|$ for some $t_0\in[0,1]$. Then $t_0\neq x_i$, $i=1,\ldots,n+r$. If $q(t_0)=0$ then Rolle's theorem implies that $q^{(n-1)}(t)$ has $r+2$ distinct zeros. Thus if the knots of q are $\xi_1,\ldots,\xi_r$ and $\xi_0=0<\xi_1<\cdots<\xi_r<1=\xi_{r+1}$ then $q^{(n-1)}(t)$ has at least two zeros in some interval $[\xi_i,\xi_{i+1}]$ where it is a linear function of slope ± 1, which is impossible. Therefore $q(t_0)\neq 0$ and

$$\frac{f(t_0)}{q(t_0)}\,q(t)$$

is a perfect spline agreeing with f at x, t_0 and by the minimal property of interpolative perfect splines

$$1<\left|\frac{f(t_0)}{q(t_0)}\right|\leqq\|f^{(n)}\|\leqq 1,$$

a contradiction which proves the lemma.

Remark 1. $q(t)\neq 0$ for $t\in[0,1]$, $t\neq x_i$ and q has precisely r knots.

Remark 2. If $n=1$ then the nodes $x_1,\ldots,x_{r+1}$ are distinct, and $\xi_j=(x_j+x_{j+1})/2$, $j=1,\ldots,r$. If $\Delta=\max(x_{j+1}-x_j)$, $j=1,\ldots,r$ then $\|q\|=\max(|q(0)|,\frac{\Delta}{2},|(q(1)|)$.

Let S_K denote a recovery scheme that assigns to each $F\in\mathscr{F}$ any one of the interpolating perfect splines of degree n whose existence is affirmed in Karlin [6]. Let q denote the "unique" perfect spline of degree n having at most r knots and satisfying (2) and (3). Then we have

Theorem 1. The positive constant $c=\|q\|$ satisfies

$$c\leqq E(x)\leqq E_{S_K}(x)\leqq 2\,c.$$

Proof. i) If S is any recovery scheme then either

$$\|q-SQ\|\geqq\|q\|=c$$

or

$$\|-q-S(-Q)\|=\|-q-SQ\|\geqq\|q\|=c,$$

by the triangle inequality. Therefore $E_S(x)\geqq c$ and $E(x)\geqq c$.

ii) Suppose $f\in W_\infty^{(n)}[0,1]$, $\|f^{(n)}\|\leqq 1$ and

$$\|S_K F-f\|=M>0.$$

Let ξ be a point of $[0,1]$ such that

$$|(S_K F)(\xi)-f(\xi)|=M.$$

Then $\xi\neq x_i$ and

$$h_\xi=\|(S_K F_\xi)^{(n)}\|\leqq\frac{1}{M}\|(S_K F-f)^{(n)}\|$$

in view of the optimality of the perfect spline interpolation with respect to the nodes (x,ξ). This same optimality with respect to the nodes x implies that

$$\|(S_K F)^{(n)}\|\leqq\|f^{(n)}\|\leqq 1,$$

and hence

$$h_\xi\leqq\frac{2}{M}.$$

Thus, according to Lemma 1, and since $S_K F_\xi(\xi)=1$,

$$c=\|q\|=\frac{\|S_K F_\xi\|}{h_\xi}\geqq\frac{M}{2},$$

and the theorem is proved.

Remark.

$$c(x)=\bar{c}(x)=\max_{\substack{f\in W_\infty^{(n)}[0,1]\\ f(x_i)=0,\ i=1,\dots,n+r}}\frac{\|f\|}{\|f^{(n)}\|}. \tag{4}$$

For, clearly $\bar{c}(x)\geqq\|q\|=c(x)$. The reverse inequality is an immediate consequence of the inequality which was used to establish Lemma 1.

Section 2

We show next by explicit construction of (linear) recovery schemes that $E(x)=c(x)$.

Theorem 2. Let $\xi_1,\dots,\xi_r$ satisfying $(\xi_0=)\ 0<\xi_1<\cdots<\xi_r<1\ (=\xi_{r+1})$, be the knots of q. Then given $f\in W_\infty^{(n)}[0,1]$ there exists a unique spline of degree $n-1$ of the form

$$s(t)=\sum_{j=0}^{n-1}a_j t^j+\sum_{j=1}^{r}b_j(t-\xi_j)_+^{n-1}, \tag{5}$$

call it s_f, which satisfies

$$s_f(x_i)=f(x_i),\qquad i=1,\dots,n+r. \tag{6}$$

Moreover, if $\|f^{(n)}\|\leqq 1$ then

$$\|f-s_f\|\leqq c(x). \tag{7}$$

Proof. Observe that if some x_i of multiplicity n coincides with a ξ_j then (6) is not well defined. We begin therefore by showing that this situation cannot occur.

Suppose then that $\xi_j = x_m = x_{m+1} = \cdots = x_{m+n-1}$.

Let

$$q_1(t) = \sum_{i=0}^{n-1} a_i^{(1)} t^i + \frac{1}{n!}\left[t^n + 2\sum_{i=1}^{m-1} (-1)^i (t-\xi^{(1)})_+^n\right]$$

be the "unique" perfect spline of degree n having $m-1$ knots and satisfying

$$q_1(x_i) = 0, \qquad i = 1, \ldots, m+n-1;$$

and let

$$q_2(t) = \sum_{i=0}^{n-1} a_i^{(2)} t^i + \frac{1}{n!}\left[t^n + 2\sum_{i=1}^{r-(m-1)} (-1)^i (t-\xi_i^{(2)})_+^n\right]$$

be the "unique" perfect spline of degree n having $r-(m-1)$ knots and satisfying

$$q_2(x_i) = 0, \qquad i = m, \ldots, n+r.$$

Note that $q_1^{(k)}(\xi_j) = q_2^{(k)}(\xi_j)$, $k = 0, 1, \ldots, n-1$. Thus by choosing the appropriate sign

$$s(t) = \begin{cases} \pm q_1(t), & 0 \leqq t \leqq \xi_j \\ \pm q_2(t), & \xi_j \leqq t \leqq 1 \end{cases}$$

is a perfect spline of degree n with r knots, $\xi_1^{(1)}, \ldots, \xi_{m-1}^{(1)}; \xi_1^{(2)}, \ldots, \xi_{r-(m-1)}^{(2)}$, none of which is ξ_j, satisfying

$$s(x_i) = 0, \qquad i = 1, \ldots, n+r$$

and

$$\|s^{(n)}\| = 1,$$

contradicting Lemma 1.

We turn now to the proof of the theorem. To establish the existence and uniqueness of s_f it suffices to show that if $s(x_i) = 0$, $i = 1, \ldots, n+r$ then $s \equiv 0$. Suppose then that $s(x_i) = 0$, $i = 1, \ldots, n+r$. Choose $\lambda \in [0, 1]$ such that $s(\lambda) \neq 0$ and $\alpha = q(\lambda)/s(\lambda)$. Thus $q(t) - \alpha s(t)$ has $n+r+1$ zeros in $[0, 1]$ and its derivative of order $n-2$, $q^{(n-2)}(t) - \alpha s^{(n-2)}(t)$ is a quadratic on each segment (ξ_i, ξ_{i+1}), $i = 0, \ldots, r$ with second derivative alternately ± 1. $g(t) = q^{(n-2)}(t) - \alpha s^{(n-2)}(t)$ has at least $r+3$ zeros, counting multiplicities, in $[0, 1]$ by Rolle's theorem, and is continuous on $[0, 1]$. We next obtain a contradiction by showing that g has at most $r+2$ zeros in $[0, 1]$. There is no loss in generality in assuming that $g''(t) = (-1)^k$ in (ξ_k, ξ_{k+1}), $k = 0, \ldots, r$, and we do so. Let $Z(g, B)$ denote the number of zeros, counting multiplicities of g in B, a subset of $[0, 1]$. We claim first: if for some k, $1 \leqq k \leqq r+1$, $Z(g, [0, \xi_k]) = k+1$, then either $g(\xi_k) = 0$ or $\operatorname{sgn} g(\xi_k) = (-1)^{k-1}$. This proposition is clearly valid for $k = 1$. Suppose it holds for $k = m$ and $Z(g, [0, \xi_{m+1}]) = m+2$. If $Z(g, [0, \xi_m]) = m+1$, then by the inductive hypothesis either $g(\xi_m) = 0$ or $\operatorname{sgn} g(\xi_m) = (-1)^{m-1}$, and since $g''(t) = (-1)^m$ in (ξ_m, ξ_{m+1}) then $g(\xi_{m+1}) = 0$ or $\operatorname{sgn} g(\xi_{m+1}) = (-1)^m$. If $Z(g, [0, \xi_m]) = m$, which is the only other possibility since $Z(g, (\xi_m, \xi_{m+1}]) \leqq 2$, then if $g(\xi_{m+1}) \neq 0$, $Z(g, (\xi_m, \xi_{m+1})) = 2$ which implies that $\operatorname{sgn} g(\xi_{m+1}) = (-1)^m$, thus completing the induction and establishing our first claim.

Now we claim that $Z(g, [0, \xi_k]) \leqq k+1$, $k=1, \ldots, r+1$. This inequality is obviously valid for $k=1$. Suppose it holds for $k=m$. If $Z(g, [0, \xi_m]) \leqq m$ then certainly $Z(g, [0, \xi_{m+1}]) \leqq m+2$. If $Z(g, [0, \xi_m]) = m+1$ then, as we have just seen, either $g(\xi_m)=0$ or $\operatorname{sgn} g(\xi_m) = (-1)^{m-1}$ and since $g''(t) = (-1)^m$ in (ξ_m, ξ_{m+1}), g has at most one zero in (ξ_m, ξ_{m+1}) so that $Z(g, [0, \xi_{m+1}]) \leqq m+2$, establishing our second claim. Now take $k=r+1$ and g has at most $r+2$ zeros in $[0, 1]$. Thus $s(\lambda)=0$ for all $\lambda \in [0, 1]$, and s_f exists and is unique.

Finally we must prove (7). If $\|f^{(n)}\| \leqq 1$ and (7) does not hold, then there exists $t_0 \in [0, 1]$, $t_0 \neq x_i$, $i=1, \ldots, n+r$, and $|f(t_0) - s_f(t_0)| > |q(t_0)|$. Thus there exists a number α, $|\alpha| > 1$ such that

$$f(t) - s_f(t) - \alpha\, q(t)$$

has $n+r+1$ zeros, and Rolle's theorem implies that

$$h(t) = f^{(n-2)}(t) - s_f^{(n-2)}(t) - \alpha\, q^{(n-2)}(t)$$

has at least $r+3$ zeros in $[0, 1]$. However, observe that since $|\alpha| > 1$, $h''(t)$ does not change sign in any interval (ξ_k, ξ_{k+1}), $k=0, \ldots, r$ and, indeed alternates in sign in these intervals. It was precisely this behavior of the function $g(t)$, not the fact that it was made up of quadratic pieces, that enabled us to conclude that g had no more than $r+2$ zeros in $[0, 1]$, and so a similar argument leads to the same conclusion for h. This contradiction implies that (7) holds and completes the proof of the theorem.

Remark. By putting $f=q$ in (7) we see that

$$\max_{\substack{f \in W_\infty^{(n)}[0,1] \\ \|f^{(n)}\| \leqq 1}} \|f - s_f\| = c(x).$$

However, this does not imply that our quest is over since the mapping $F \to s_f$ is not a recovery scheme, for it produces an element of $W_\infty^{(n-1)}[0, 1]$, not necessarily a function in $W_\infty^{(n)}[0, 1]$ as required. Our next step, therefore, is to smooth s_f a little, at least in the case $n > 1$.

Suppose $n \geqq 2$. If $g \in C[0, 1]$ let $V_m(g)$ be the de La Vallée Poussin means of g with respect to the Chebyshev polynomials for the interval $[0, 1]$. (cf. de La Vallée Poussin [5]). $V_m(g) = g$ if g is a polynomial of degree $\leqq m$, $V_m(g)$ is a polynomial of degree at most $2m-1$ for any $g \in C[0, 1]$, and

$$\|V_m(g) - g\| \leqq 4\, E_m(g) \tag{8}$$

where $E_m(g)$ is the error of the best approximation to g by a polynomial of degree at most m.

Choose $m \geqq n-1$ and consider the recovery scheme $F \to V_m(s_f)$. Then

$$\begin{aligned} c(x) \leqq E(x) &\leqq \sup_{\substack{f \in W_\infty^{(n)}[0,1] \\ \|f^{(n)}\| \leqq 1}} \|f - V_m(s_f)\| \\ &\leqq \sup_{\substack{f \in W_\infty^{(n)}[0,1] \\ \|f^{(n)}\| \leqq 1}} \|f - s_f + s_f - V_m(s_f)\| \\ &\leqq c(x) + \sup_{\substack{f \in W_\infty^{(n)}[0,1] \\ \|f^{(n)}\| \leqq 1}} \|s_f - V_m(s_f)\|, \end{aligned}$$

in view of (7). Since both s_f and $V_m(s_f)$ preserve polynomials of degree at most $n-1$, Peano's theorem yields

$$s_f(\tau)-V_m(s_f,\tau)=\frac{1}{(n-1)!}\int_0^1 [s_h(\tau)-V_m(s_h,\tau)]\, f^{(n)}(t)\, d\,t \tag{9}$$

where $h(u,t)=(u-t)_+^{n-1}$, and so

$$s_h(\tau)=\sum_{i=1}^{n+r}(x_i-t)_+^{n-1}L_i(\tau)$$

where $L_i(\tau)$ are splines of the form (5) satisfying $L_i(x_j)=\delta_{ij}$, with the usual modifications for coalescent nodes. Suppose

$$|s_f(\bar\tau)-V_m(s_f,\bar\tau)|=\|s_f-V_m(s_f)\|.$$

Then $g(t)=s_h(\bar\tau)$ has modulus of continuity $\omega(g,\delta)\leqq C_1\,\delta^{n-1}$, for some constant C_1. Thus (8) and (9) yield, for $\|f^{(n)}\|\leqq 1$,

$$\|s_f-V_m(s_f)\|\leqq C_2\,E_m(g)\leqq\frac{C_3}{m^{n-1}},$$

the last inequality following from Jackson's theorem. We have now proved the following.

Theorem 3. Given $\varepsilon>0$ there exists M such that for $m>M$

$$E_{V_m(s_f)}<c(x)+\varepsilon,$$

provided that $n\geqq 2$.

A separate treatment is required for the case $n=1$, and in this case an optimal algorithm is obtained.

Theorem 4. If $n=1$ let S_1F be the polygonal line connecting the points $(x_i, f(x_i))$, $i=1,\ldots,r+1$ (the x_i are necessarily distinct), defined to be $f(x_1)$ in $0\leqq t\leqq x_1$ and $f(x_{r+1})$ in $x_{r+1}\leqq t\leqq 1$, then

$$\sup_{\substack{f\in W_\infty^{(1)}[0,1]\\ \|f'\|\leqq 1}}\|f-S_1F\|=c(x)$$

Proof. Consider the interval $[x_i, x_{i+1}]$ where $1\leqq i\leqq r$. On this interval the segment connecting $(x_i, f(x_i))$ to $(x_{i+1}, f(x_{i+1}))$ is (in non-trivial cases) the diagonal of a rectangle whose sides, issuing from $(x_i, f(x_i))$, have slopes 1 and -1 respectively. If $\|f'\|\leqq 1$ then $(t, f(t))$ is in this rectangle for $x_i\leqq t\leqq x_{i+1}$, and an easy computation yields, for this interval,

$$|f(t)-(S_1F)(t)|\leqq\frac{x_{i+1}-x_i}{2}\left[1-\left(\frac{f(x_{i+1})-f(x_i)}{x_{i+1}-x_i}\right)^2\right],$$

which implies $|f(t)-(S_1F)(t)|\leqq\frac{\Delta}{2}$, $x_1\leqq t\leqq x_{r+1}$, where $\Delta=\max(x_{i+1}-x_i)$, $i=1,\ldots,r$.

It is clear that in $[x_{r+1}, 1]$, $|f(t)-(S_1F)(t)|\leqq 1-x_{r+1}$ while in $[0, x_1]$ $|f(t)-(S_1F)(t)|\leqq x_1$. The theorem now follows from Remark 2 following Lemma 1 and the definition of c.

Remark. The recovery schemes given in Theorems 3 and 4 which are almost optimal and optimal, respectively, are all linear mappings from F into $W_\infty^{(n)}[0, 1]$. They reveal that $E(x)=c$.

Section 3

This section is devoted to estimates of $c(x)$ in terms of x and n. We begin with some information about $c(x)$ as a function of x and of the interval in which the problem is set.

Suppose $x:(x_1, \ldots, x_{n+r})$ and $y:(y_1, \ldots, y_{n+s})$ satisfy $0 \leqq x_1 \leqq \cdots \leqq x_{n+r} \leqq 1$ with $x_i < x_{i+n}$, $i=1, \ldots, r$, and $0 \leqq y_1 \leqq \cdots \leqq y_{n+s} \leqq 1$ with $y_i < y_{i+n}$, $i=1, \ldots, s$; and each point of y is a point of x appearing in x with multiplicity at least that with which it appears in y, then we say that x is a refinement of y and write $x < y$.

Lemma 2. If $x < y$ then $c(x) \leqq c(y)$.

Proof. Follows immediately from (4).

Let I denote the interval $[0, 1]$ then since the quantity $c(x)$ depends on I we write it as $c(x, I)$. If J is any closed interval in I and x a set of $n+r$ nodes in J we may define, as above, $c(x, J)$. Suppose we are given nodes $x \in I$ and

$$0 < x_j = x_{j+1} = \cdots = x_{j+n-1} < 1,$$

i.e. x_j is an interior node of maximal multiplicity. Then we have

Lemma 3.

$$c(x, I) = \max\{c(x^{(1)}, [0, x_j]), c(x^{(2)}, [x_j, 1])\}$$

where $x^{(1)}$ is $(x_1, \ldots, x_{j+n-1})$ and $x^{(2)}$ is $(x_j, \ldots, x_{n+r})$.

Proof. i) Suppose q is the perfect spline of Lemma 1. Put

$$f_1 = \begin{cases} q, & 0 \leqq t \leqq x_j \\ 0, & x_j < t \leqq 1 \end{cases}; \qquad f_2 = \begin{cases} 0, & 0 \leqq t \leqq x_j \\ q, & x_j \leqq t < 1 \end{cases}$$

then $q = f_1 + f_2$ and $f_1 \in W_\infty^{(n)}[0, x_j]$, $f_2 \in W_\infty^{(n)}[x_j, 1]$. Let $\|\cdot\|_i$, $i=1, 2$ be the essential sup norm for the intervals $[0, x_j]$ and $[x_j, 1]$ respectively. Then

$$\|f_1^{(n)}\|_1 = \|f_2^{(n)}\|_2 = 1$$

and

$$\begin{aligned} c(x, I) = \|q\| &= \max(\|f_1\|_1, \|f_2\|_2) \\ &\leqq \max\{c(x^{(1)}, [0, x_j]), c(x^{(2)}, [x_j; 1])\}. \end{aligned} \tag{10}$$

ii) Suppose $f \in W_\infty^{(n)}[0, x_j]$ satisfies $\|f^{(n)}\|_1 = 1$ and $f(x_i) = 0$, $i=1, \ldots, j+n-1$. Put

$$h(x) = \begin{cases} f(x), & 0 \leqq x \leqq x_j \\ 0, & x_j < x \leqq 1. \end{cases}$$

Then $h \in W_\infty^{(n)}[0, 1]$, $\|h^{(n)}\| = 1$ and $h(x_i) = 0$, $i=1, \ldots, n+r$. Thus

$$\|f\|_1 = \|h\| \leqq c(x, I)$$

and $c(x^{(1)}, [0, x_j]) \leqq c(x, I)$. A similar argument yields $c(x^{(2)}, [x_j, 1]) \leqq c(x, I)$ and the Lemma is proved.

Remark. The condition that x_j have maximal multiplicity is not used in establishing (10) which, therefore, holds for arbitrary x.

Theorem 5. i) $c(x) \leqq \frac{1}{n!}$.

ii) If $x_1 = 0$, $x_{n+r} = 1$ and $\Delta = \max(x_{i+1} - x_i)$, $i = 1, \ldots, n+r-1$ then,

$$c(x) \leqq \frac{(n-1)^{n-1}}{n^n} \frac{1}{n!}, \tag{11}$$

$$c(x) \leqq \frac{(n-1)^{n-1}}{n!} \left(\frac{\Delta}{2}\right)^n, \tag{12}$$

and

$$c(x) \leqq \Delta^n. \tag{13}$$

Proof. Let $|q(\bar{t})| = \|q\|$. Suppose $x_m < \bar{t} < x_{m+1}$. Consider the divided difference $q(\bar{t}, x_i, \ldots, x_{i+n-1})$ where $1 \leqq i \leqq m \leqq i+n-2$. Then we have

$$\frac{c(x)}{\prod_{j=0}^{n-1} |\bar{t} - x_{i+j}|} = \frac{|q(\bar{t})|}{\prod_{j=0}^{n-1} |\bar{t} - x_{i+j}|} = |q(\bar{t}, x_i, \ldots, x_{i+n-1})| \leqq \frac{\|q^{(n)}\|}{n!} = \frac{1}{n!} \tag{14}$$

which yields i) immediately. (11) and (12) follow from inequalities of Beesack [1, p. 808] and Das and Vatsala [3, Lemma 4.1]. Since

$$\bar{t} - x_{m-j} \leqq (j+1)\Delta,\ j = 0, 1, \ldots, m-i \text{ and } x_{m+j} - \bar{t} \leqq j\Delta,\ j = 1, \ldots, i+n-1-m,$$

$$\prod_{j=0}^{n-1} |\bar{t} - x_{i+j}| \leqq (m+1-i)!\,(n-(m+1-i))!\,\Delta^n \leqq n!\,\Delta^n$$

and (13) follows from (14).

Remark. A referee has observed that it is easy to show that $c(x) \leqq (\Delta^n)/(4n)$, which for $n > 3$ is an improvement over (12) and (13).

Louboutin (cf. Schoenberg [8, p. 8]) showed that there exists a perfect spline of degree $(n+1)$, $f(x)$, which satisfies

$$f^{(k)}(-1) = 0, \qquad k = 0, \ldots, n$$

$$f(1) = 1,\ f^{(k)}(1) = 0, \qquad k = 1, \ldots, n$$

and

$$\max_{-1 \leqq t \leqq 1} |f^{(n+1)}(t)| = 2^{n-1} n!.$$

As Schoenberg observes,

$$f'(t) = M(t; \eta_0, \ldots, \eta_{n+1}),$$

where

$$\eta_j = -\cos \frac{j\pi}{n+1}, \qquad j = 0, \ldots, n+1$$

and $M(t; \eta_0, \ldots, \eta_{n+1})$ (a perfect B-spline) is the divided difference of

$$M(t; u) = (n+1)(u-t)_+^n.$$

Let

$$g(t)=\frac{M(t;\eta_0,\ldots,\eta_{n+1})}{2^{n-1}n!} \tag{15}$$

then

$$\max_{-1\leqq t\leqq 1}|g^{(n)}(t)|=1,$$

$$g^{(k)}(-1)=g^{(k)}(1)=0, \qquad k=0,\ldots,n-1,$$

and

$$\mu=\max_{-1\leqq t\leqq 1}|g(t)|=\frac{M(0;\eta_0,\ldots,\eta_{n+1})}{2^{n-1}n!}\leqq\frac{n+1}{2^n n!} \tag{16}$$

where we use the fact that g attains its maximum at $t=0$, and the B-spline is bounded by $(n+1)/2$ at 0. (cf. Schoenberg [7, p. 289]).

We now have

Theorem 6. Suppose $r=m\,n$ and $x_{jn+s}=x_{jn+1}$, $s=1,\ldots,n$; $j=0,1,\ldots,m$, $x_1=0$ and $x_{n+r}=1$. Let $\Delta=\max(x_{i+1}-x_i)$, $i=1,\ldots,n+r-1$ then

$$c(x)=\mu\left(\frac{\Delta}{2}\right)^n \tag{17}$$

where μ is defined in (16).

Proof. Let $z_j=x_{jn+1}$, $j=0,1,\ldots,m$ and $z_{j+1}-z_j=\Delta_j$, $j=0,\ldots,m-1$. Put

$$g_j(u)=g\left(\frac{2(u-y_j)}{\Delta_j}\right)=g(t)$$

where $y_j=(z_j+z_{j+1})/2$ and $z_j\leqq u\leqq z_{j+1}$. Then g_j is a perfect spline on $[z_j,z_{j+1}]$ with n interior knots,

$$g_j^{(k)}(z_j)=g_j^{(k)}(z_{j+1})=0,\ k=0,1,\ldots,n-1,$$

and

$$\max_{z_j\leqq u\leqq z_{j+1}}|g_j^{(n)}(u)|=\left(\frac{2}{\Delta_j}\right)^n.$$

Then, in the notation of Lemma 3,

$$\begin{aligned}c(z^{(j)},[z_j,z_{j+1}])&=\max_{z_j\leqq u\leqq z_{j+1}}|g_j(u)|\left(\frac{\Delta_j}{2}\right)^n\\&=\mu\left(\frac{\Delta_j}{2}\right)^n,\end{aligned}$$

and successive applications of Lemma 3 yield (17).

Remark. Let $z:(z_1,\ldots,z_{n+r})$ be any set of nodes, then in view of Lemma 2

$$c(z)>\mu\left(\frac{\Delta}{2}\right)^n\geqq\frac{\Delta^n}{4^n n!}.$$

The final inequality follows from the simple observation that

$$M(0;\eta_0,\ldots,\eta_{n+1})\geqq 1/2.$$

We conclude by observing that there is a set of nodes, x^*, satisfying $c(x^*)\leqq c(x)$ for all x. $c(x^*)$ is the $n+r$ width of $W_\infty^{(n)}[0,1]$ in the sense of Gelfand. It probably comes as no surprise that the nodes x^* are the zeros of the *Chebyshev perfect spline* of degree n with r knots of the interval $[0,1]$. For details see Tihomirov [9].

Postscript

After this paper was completed we became acquainted with related work of Gaffney and Powell (Optimal Interpolation, C. S. S. 16, Computer Science and Systems Division, A. E. R. E., Harwell, Oxfordshire, England, May, 1975) and Bojanov (Optimal Methods of Interpolation in $W^{(r)}L_q(M, a, b)$, Comptes rendus de l'Académie bulgare des Sciences, Tome 27, 1974, pp. 885–888.) Gaffney and Powell show, by other methods, that the spline interpolation described in Theorem 2 provides optimal recovery of the value $f(t)$, t fixed, $0 \leqq t \leqq 1$, $t \neq x_i$, $i = 1, \ldots, n+r$. Note that this result, as well as the "optimality" of s_f in recovering, say, $f'(t)$ for fixed t, with $t = x_i$ or $t \in [0, 1] \backslash (x_1, x_{n+r})$ (among other linear functionals) is tacit in the proofs of Theorems 1 and 2 in view of the pointwise inequalities established there and the reliance on Rolle's theorem.

References

1. Beesack, P. R.: On the Green's function of an N-point boundary value problem. Pacific J. Math., **12**, 801–812 (1962)
2. Cavaretta, A. S. Jr.: Oscillation and zero properties for perfect splines and monosplines. J. d'Analyse Math. **28**, 41–59 (1975)
3. Das, K. M., Vatsala, A. S.: On Green's function of an n-point boundary value problem. Trans. A.M.S., **182**, 469–480 (1973)
4. de Boor, C.: A remark concerning perfect splines. Bull. A.M.S., **80**, 724–727 (1974)
5. de La Vallée Poussin, C. J.: Leçons sur l'Approximation des Fonctions d'une Variable Réele. Paris: Gauthier-Villars 1952
6. Karlin, S.: Some variational problems on certain Sobolev spaces and perfect splines, Bull. A.M.S., **79**, 124–128 (1973). Also, Trans. A.M.S., **106**, 25–66 (1975)
7. Schoenberg, I. J.: On spline functions, Inequalities. Academic Press, 1967, p. 255–291
8. Schoenberg, I. J.: Cardinal Spline Interpolation. S.I.A.M. Philadelphia, 1973
9. Tihomirov, V. M.: Best methods of approximation and interpolation of differentiable functions in the space $C_{[-1,1]}$. Math. USSR Sbornik, **9**, 275–289 (1969)

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